

Save As, Copy Figure, Print, Show Code, FILE

TITLE AND LABELS

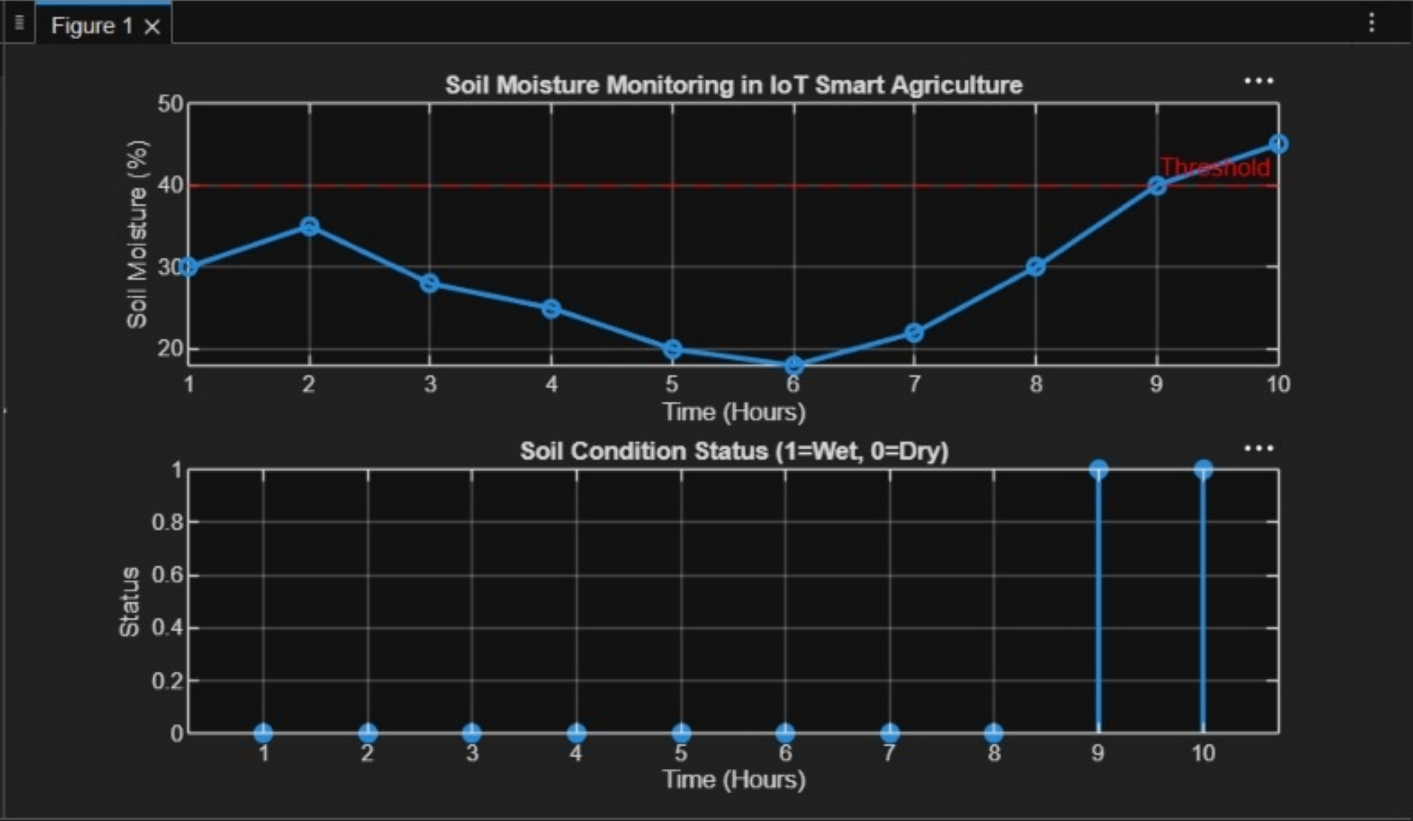
Title Subtitle X-Label Y-Label Z-Label Legend Colorbar

GRID

Grid X-Grid Y-Grid

LABELS AND ANNOTATIONS

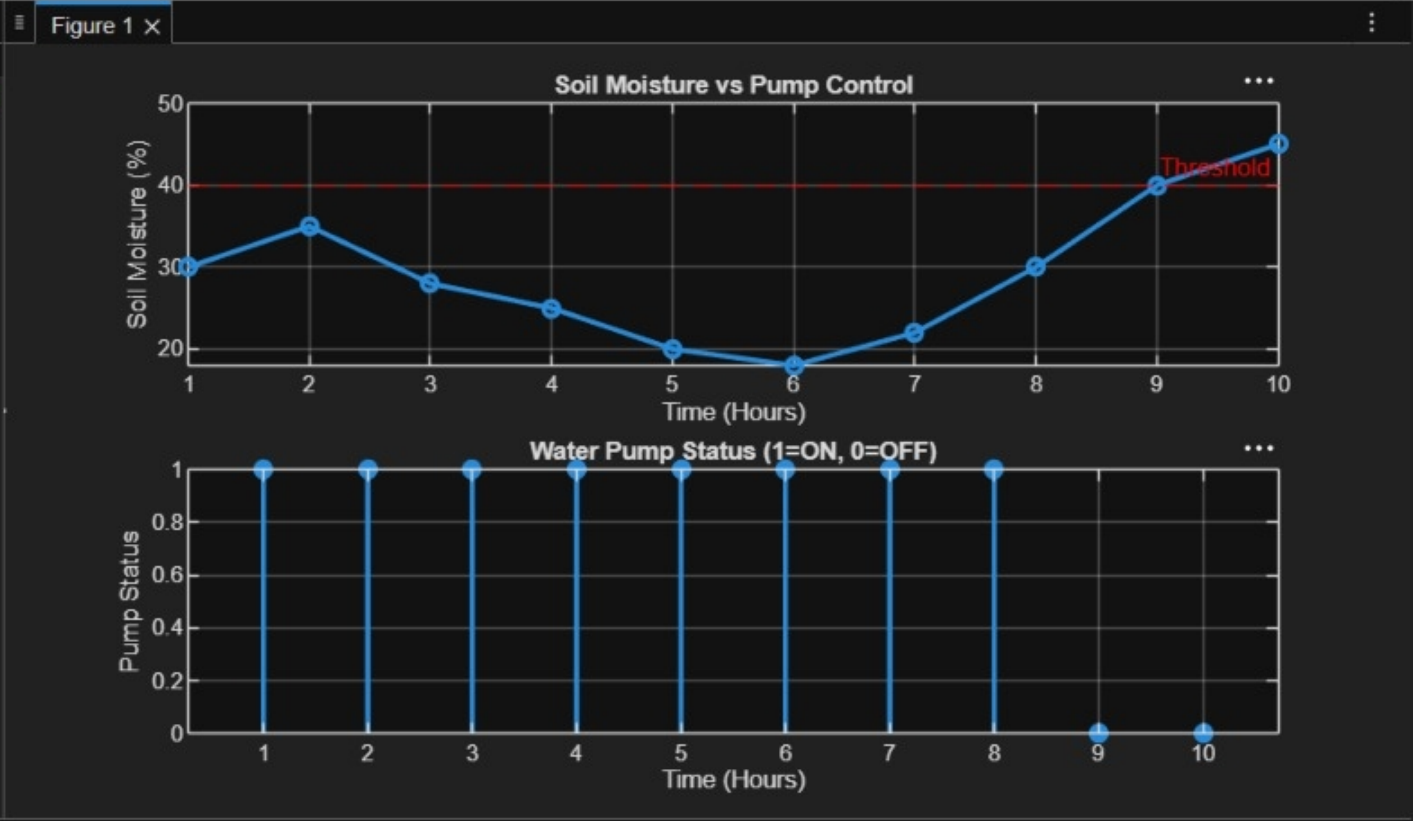
```
untitled30.m x untitled31.m x +
/MATLAB Drive/untitled31.m
4 soil_moisture = [30 35 28 25 20 18 22 30 40 45];
5 % Threshold for irrigation
6 threshold = 40;
7 figure
8 % ----- Graph 1: Soil Moisture Variation -----
9 subplot(2,1,1)
10 plot(time, soil_moisture, '-o', 'LineWidth', 2)
11 title('Soil Moisture Monitoring in IoT Smart Agriculture')
12 xlabel('Time (Hours)')
13 ylabel('Soil Moisture (%)')
14 grid on
15 hold on
16 yline(threshold, 'r--', 'Threshold')
17 % ----- Graph 2: Moisture Status (Dry/Wet Level) -----
18 status = soil_moisture >= threshold;
19 subplot(2,1,2)
20 stem(time, status, 'filled', 'LineWidth', 2)
21 title('Soil Condition Status (1=Wet, 0=Dry)')
22 xlabel('Time (Hours)')
23 ylabel('Status')
24 grid on
25
26
```



Copy Figure, Print, Show Code, FILE, LABELS AND ANNOTATIONS

Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

```
3 time = 1:10;
4 soil_moisture = [30 35 28 25 20 18 22 30 40 45];
5 % Threshold level
6 threshold = 40;
7 % Pump control logic (1 = ON, 0 = OFF)
8 pump_status = soil_moisture < threshold;
9 figure
10 % ----- Graph 1: Soil Moisture vs Pump Status -----
11 subplot(2,1,1)
12 plot(time, soil_moisture, '-o','LineWidth',2)
13 title('Soil Moisture vs Pump Control')
14 xlabel('Time (Hours)')
15 ylabel('Soil Moisture (%)')
16 grid on
17 hold on
18 yline(threshold,'r--','Threshold')
19 % ----- Graph 2: Pump Status (ON/OFF) -----
20 subplot(2,1,2)
21 stem(time, pump_status, 'filled','LineWidth',2)
22 title('Water Pump Status (1=ON, 0=OFF)')
23 xlabel('Time (Hours)')
24 ylabel('Pump Status')
25 grid on
```



Save As Copy Figure Print Show Code

FILE

Labels and Annotations: Title Subtitle X-Label Y-Label Z-Label Legend Colorbar Grid X-Grid Y-Grid

```
3 time = 1:10;
4 soil_moisture = [30 35 28 25 20 18 22 30 40 45];
5 % Threshold level
6 threshold = 40;
7 % Irrigation efficiency calculation (%)
8 efficiency = (soil_moisture ./ threshold) * 100;
9 efficiency(efficiency > 100) = 100; % limit to 100%
10 figure
11 % ----- Graph 1: Soil Moisture vs Efficiency -----
12 subplot(2,1,1)
13 plot(time, soil_moisture, '-o','LineWidth',2)
14 title('Soil Moisture Variation')
15 xlabel('Time (Hours)')
16 ylabel('Soil Moisture (%)')
17 grid on
18 hold on
19 yline(threshold,'r--','Threshold')
20 % ----- Graph 2: Irrigation Efficiency -----
21 subplot(2,1,2)
22 plot(time, efficiency, '-s','LineWidth',2)
23 title('Irrigation Efficiency (%) in Smart Agriculture')
24 xlabel('Time (Hours)')
25 ylabel('Efficiency (%)')
```

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